ANTENATAL CARE

BY

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Definition of antenatal care

* Refers to the care that is given to an expectant mother from time of conception is confirmed until the beginning of labor.
* OR:-
* Is a program of preventive obstetrics care with a main objectives to ensure a safe motherhood with a safe delivery of a healthy fetus

Aim/purpose from antenatal care

- To reduce maternal and perinatal mortality and morbidity rates.
- To improve the physical and mental health of women and children.
- To prepare the woman for labor, lactation, and care of her infant.
- To detect early and treat properly complicated conditions that could endanger the life or impair the health of the mother or the fetus.

Terms Used in Describing a Pregnant Client

* Abortion Loss of pregnancy before the age of viability (20 weeks gestation)
* GP/TPAL Gravida, para
* TPAL (Term, Preterm, Abortions, living children

* Gravida Pregnancy, regardless of duration, includes present pregnancy
* Para :Delivery (birth) after 20 weeks'gestation, whether infant born alive or dead.
* Preterm Delivery after 20 weeks' gestation but before 38 weeks (full term)
* Term A pregnancy between 38 and 42 weeks' gestation

* Nulligravida Never been pregnant
* Primigravida Pregnant for first time
* Multigravida Pregnant two or more times
* Nullipara Never having delivered an infant after
* 20 weeks' gestation
* Primipara Has delivered once after 20 weeks' gestation
* Multipara Has delivered twice or more e after 20 weeks' gestation
* Postterm Delivery after 42 weeks' gestation

Pre-conception visit

* The ideal first visit should be at pre-conception where health education and risk assessment can be directed towards the planned pregnancy in order to allow the pregnancy to be started in the optimum conditions.

Services offered to pregnant women

1· Screening and risk assessment:

* History taking.
* Examination of mother.
* Laboratory Investigations.

2-- Immunization against tetanus.

3-- Early recognition, management of those who have developed complications.

4- Health Education;

Screening and risk assessment:

Frequency of visits:

Frequency of visits:

- ***Low risk patients.*** **every month till 28 weeks then every 2 weeks till 36 week then weekly till the end of pregnancy.**
- ***High risk patients.*** **every 2 weeks till 28 weeks then weekly till 36 week then hospitalization or according to the condition.**

1st visit (Booking visit):

Clinical evaluation including:

Complete history taking

. General examination

Abdominal examination

. Local examination

★ *Subsequent visits:*

* *Clinical evaluation :-*

* *screening tests* as:

- Complete blood picture.

- Urine analysis for glucose or proteinuria.

- Measuring fundal height
- * Hearing FHS , measuring BP, weight gain , fetal movement,

History

- Welcome the woman, and ensure a quite place where she can express concerns and anxiety without being overheard by other people.

- Personal and social history (Demography):

This include: woman's name, age, occupation ,address , year of marriage, and phone number.

Family history

Family history provides valuable information about the general health of the family, and it may reveal information about patters of genetic or congenital anomalies.

Including:

- D.M.
- Hypertension.
- Heart disease.
- Cancer.
- Anemia.
- Others.

Medical and surgical history

past attack or present

Chronic condition such as

- Diabetes mellitus
- Hypertension
- And renal disease
- others

Menstrual History

Menstrual history is important to establish the estimated date of delivery(EDD OR EDC).It includes:

- Age of menarche.
- Last menstrual period (LMP).
- Regularity ,duration ,and frequency of menstrual cycle.
- Contraception method.
- Expected date of delivery (EDD) is calculated as followed:

1st day of LMP −3 months +7 days, and change the year.

Example:. LMP: 15/7/2011

PRESENT HISTORY

Ask the patient if she has any current problem, such as:

- Nausea & vomiting.
- Abdominal pain.
- Headache.
- Urinary complaints.
- Vaginal bleeding.
- Edema.
- Backache.
- Heartburn.
- Constipation.

Obstetrical history

This provides essential information about the previous pregnancies that may alert the care provider to possible problems in the present pregnancy. Which includes:

- Gravida, para, abortion, and living children ; **OB.digit**,(TPAL)(GP)(GTPALP).
- Weight of infant at birth & length of gestation.
- Mode of delivery, location of birth, and type of anesthesia.
- Maternal or infant complications.

Physical Examination

It is important :

*To detect previously undiagnosed physical problems that may affect the pregnancy outcome.

* To establish baseline levels that will guide the treatment of the expectant mother and fetus throughout pregnancy.

Vital Signs

1. Blood pressure:

It is taken to ascertain normality and provide a baseline reading for a comparison throughout the pregnancy.

2. Pulse:

The normal pulse rate = 60-90 BPM.

Tachycardia is associated with anxiety, hyperthyroidism, or infection.

3. Respiration:

The normal is 16-24 BPM.

*Tachypnea may indicate respiratory infection, or cardiac disease.

4. Temperature:

The normal during pregnancy is 36.2C to 37.6C.

*Increased temperature suggests infection.

Cardiovascular system

- **Venous congestion(DVT):**

  Can develop into varicosities, venous congestion most commonly noted in the legs, vulva, and rectum.

- **Edema:**

  Edema of the extremities or face necessitates further assessment for signs of pregnancy-induced hypertension.

Musculoskeletal system

- **Posture and gait (Lordosis)**

  Body mechanics and changes in posture and gait should be addressed. Body mechanics during pregnancy may produce strain on the muscles of the lower back and legs (**Waddling**).

Height & weight

An initial weight is needed to establish a baseline for weight gain throughout pregnancy.

Recommendation for weight gain during pregnancy are often made based on the woman’s body mass index (about 12 kg, at the end of pregnancy)

★ Components of increased weight:

Fetus (3.5 kgm), placenta (0.5 kgm), uterine growth (0.5 kgm), amniotic fluid (1 kgm), breast growth (0 .5 kgm), blood volume(l.5 kgm), protein retention (2 kgm) and water retention (2kgm)

Pelvic measurement

The bony pelvis is evaluated early in the pregnancy to determine whether the diameters are adequate to permit vaginal delivery.

Abdominal Examination

- **The abdomen is inspected for: skin changes , and size**

- The height of the fundus is measured , which determines the period of the gestation.

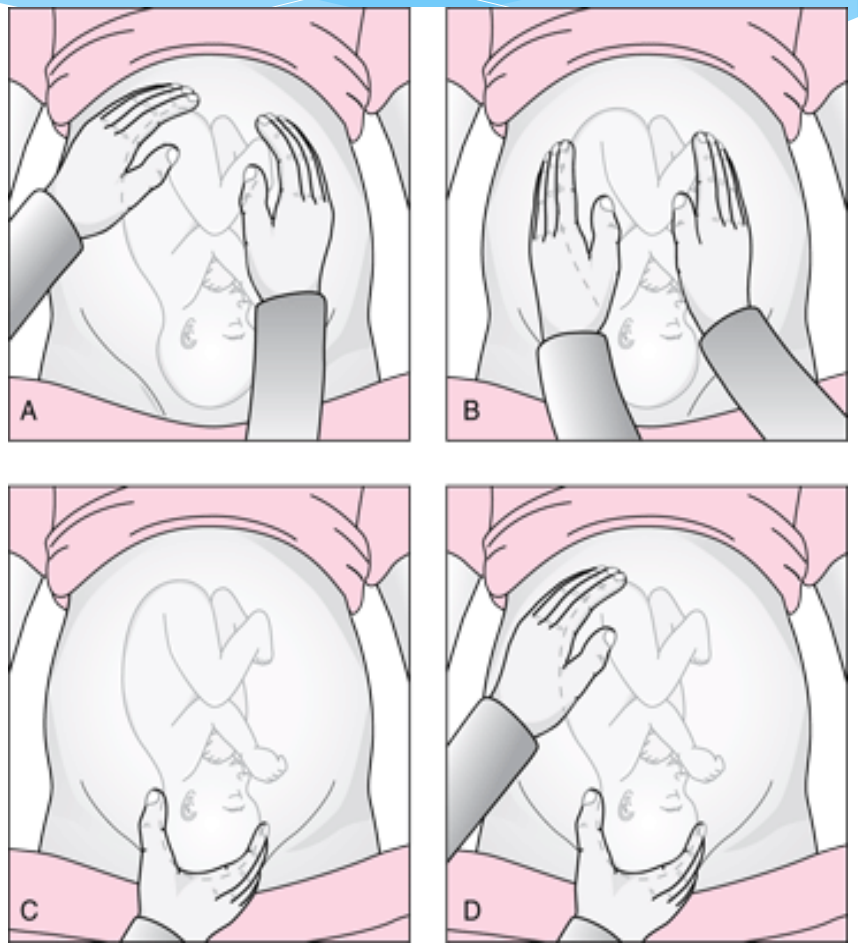
Fetal lie , attitude , presentation & position

fetal lie & position.

- the abdomen is longer if the fetal lie is longitudinal as occurs in 99.5% of cases.
- the abdomen is lower & broad if the lie is transverse.

- fetal movement is inspected as evidence of fetal life and position.

Leopold Maneuver



Fetal Heart Sound (FHS)

- fetal heart beat can be heard by stethoscope after the 20th week,
- or Doppler after 8th week.
- Normal fetal heart rate is 120-160 beats/min.

Neurological system

DTR

- Deep tendon reflexes (DTR) should be evaluated because hyperreflexia is associated with complications of pregnancy.

Skin

- Chloasma and linea nigra related to pregnancy.
- Striae graviderum should be noted.
- Nail beds should be pink with instant capillary return.
- Pallor of the skin my indicate anemia.

Neck

* The thyroid enlarges slightly during the 2nd trimester. But, gross enlargement or tenderness may indicate hyperthyrodism

Breast changes

Assess breast size, symmetry, condition of nipple, and the presence of colostrum.

GIT

- Mouth:

The gum may be red, tender, edematous as a result of the effects of increased estrogen.

- Intestine:

Assess for the bowel sound.

Assess for constipation or diarrhea.

Laboratory Examinations

TEST	Purpose
* Blood group	To determine blood type.
* Hgb & Hct	To detect anemia
* Rubella	To determine immunity
* Urine analysis	To detect infection or renal disease.
* Papanicolaou (pap) test	To screen for cervical cancer
* Chlamydia	To detect sexual transmitted disease.
* Glucose	To screen for gestational diabetes.

Sonography

Is used to

- estimate the gestational age.
- Check amniotic fluid volume.
- Check the position of the placenta.
- Detect the multifetal pregnancy.
- The position of the baby.

HEALTH EDUCATION DURING PREGNANCY

★ Antenatal advice ★

-C alories : the total calories is about 2500 Cal/day

- Proteins: during pregnancy, protein needs is about 100 gm/day and majority must be 1st class containing essential amino acids (animal sources)

- Fats: usually available in balanced diet (equal 100 mg/day)

- Carbohydrates: pregnancy need ± 300 gm/day

Hygiene

- Warm shower or sponge baths is better than tub bath.
- Hot bath should be avoided because they may cause fatigue.
- Regular washing for genital area, axilla, and breast due to increased discharge and sweating.
- Vaginal douches should avoided .

Breast care

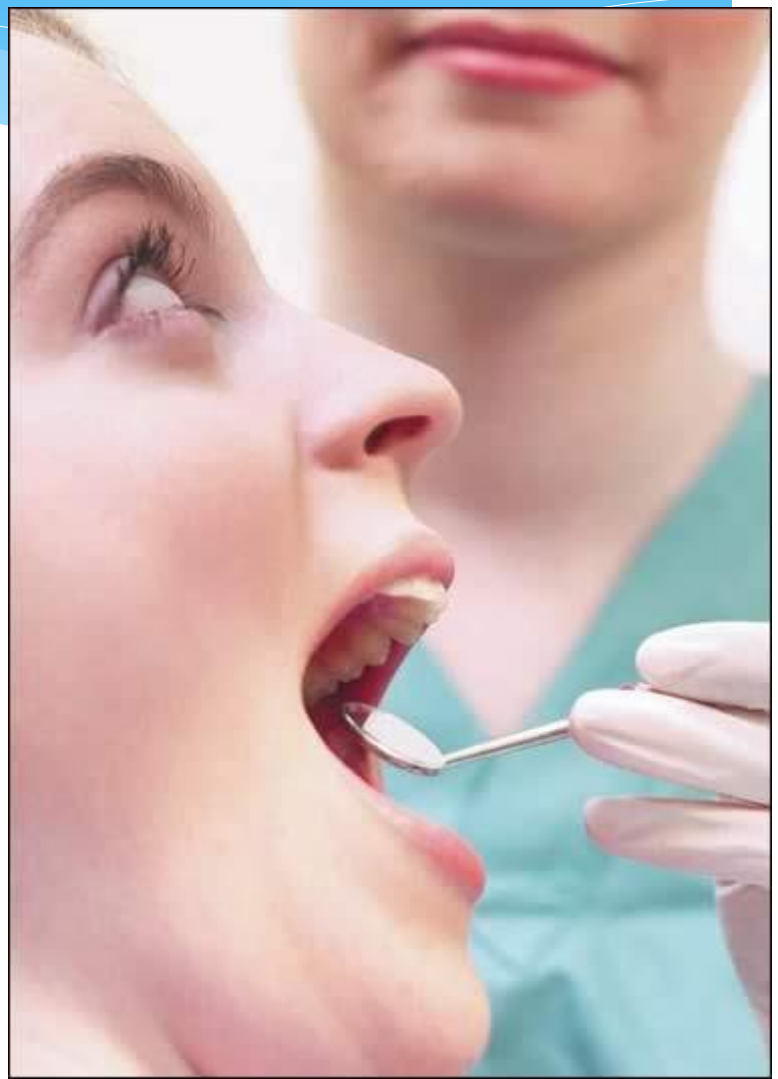
- Wear firm, supportive bra with wide straps to spread weight across the shoulder.

- Wash breasts with clean tap water (no soap, because that could be drying). Daily to remove the colostrum & reduce the risk of infection.

- It is not recommended to massage the breast, this may stimulate oxytocin hormone secretion and possibly lead to contraction.

Dental care

- The teeth should be brushed carefully in the morning and after every meal.
- Encourage the woman to see her dentist regularly for routine examination & cleaning.
- Encourage the woman to snack foods, such as fresh fruit & vegetables to avoid sugar coming in contact with the teeth.
- A tooth can be extracted during pregnancy, but local anesthesia is recommended.



Clothes

Woman should avoid wearing tight cloths such as belt or constricting bans on the legs, because these could impede lower extremity circulation.

Suggest wearing shoes with a moderate to low heel to minimize pelvic tilt & possible backache.

Loose, and light clothes are the most comfortable.

Sexual activity

- Sexual intercourse is allowed with moderation, is absolutely safe and normal unless specific problem exist such as: vaginal bleeding or ruptured membrane.

- If a woman has a history of abortion, she should avoid sexual intercourse in the early months of pregnancy.

Bowel care:

constipation is a common problem
Management is usually including:-
- High fiber diet + fluids
- Physical activities + regular bowel habit .

Exercises

*Exercise should be simple. Walking is ideal, but long period of walking should be avoided.

*The pregnant woman should avoid lifting heavy weights such as: mattresses furniture, as it may lead to abortion.

*She should avoid long period of standing because it predisposes her to varicose vein.

*She should avoid setting with legs crossed because it will impede circulation.

Sleep & Rest

- The pregnant woman should lie down to relax or sleep for 1 or 2 hours during the afternoon.
- At least 8 hours sleep should be obtained every night & increased towards term, because the highest level of growth hormone secretion occurs at sleep.
- Advise woman to use natural sedatives such as: warm bath & glass of worm milk.
- A good sleeping position is sims' position, with the top leg forward. This puts the weight of the fetus on the bed, not on the woman, and allows good circulation in the lower extremities.
- avoid resting in supine position, as supine hypotension syndrome can develop.

Travel

Many women have questions about travel during pregnancy.

- Early in normal pregnancy, there are no restrictions.
- Late in pregnancy, travel plans should take into consideration the possibility of early labor.
- Long distances car or train used is not preferred unless there is a rest in between.

Warning signs

* Vaginal bleeding
* Blurred vision
* Sever headache
* Persistent vomiting
* Generalized edema
* Sudden gush of fluid
* Abdominal pain.

Thank you